Contents

1 Basic Concepts of Lie Algebra

1.1 Definitions and first examples

Definition 1.1.1

A vector space L over a field $\mathbb{F}$, with an operation $L \times L \to L$, denoted $(x, y) \mapsto [xy]$ and called the ***bracket*** or ***commutator*** of x and y, is called a ***Lie algebra*** over $\mathbb{F}$ if the following axioms are satisfied:
(L1) The bracket operation is bilinear.
(L2) $[xx] = 0$ for all $x \in L$.
(L3) $[x[yz]] + [y[zx]] + [z[xy]] = 0 \quad (x, y, z \in L)$.

Remark.
- (L1) and (L2), applied to $[x + y, x + y]$ imply anticommutativity : (L2′) $[xy] = [-yx]$.
- Conversely, if char $\mathbb{F} \neq 2$, it is clear that (L2′) will imply (L2).

1.2 Lie algebras of derivations

Definition 1.2.1

- By an ***F-algebra*** (not necessarily associative) we simply mean a vector space $\mathfrak{A}$ over F endowed with a bilinear operation $\mathfrak{A} \times \mathfrak{A} \to \mathfrak{A}$, usually denoted by juxtaposition (unless $\mathfrak{A}$ is a Lie algebra, in which case we always use the bracket).
- By a ***derivation*** of $\mathfrak{A}$ we mean a linear map $\delta : \mathfrak{A} \to \mathfrak{A}$ satisfying the familiar product rule $\delta(ab) = a\delta(b) + \delta(a)b$ for each $n \in \mathbb{Z}$.

1.3 Ideals and homomorphism

1.3.1 Ideals

Definition 1.3.1

A subspace I of a Lie algebra L is called an ***ideal*** of L if $x \in L, y \in I$ together imply $[xy] \in I$.

1.3.2 Homomorphisms and representations

Definition 1.3.2

- A linear transformation $\phi : L \to L'$ (L, L' Lie algebras over F) is called a ***homomorphism*** if $\phi([xy]) = [\phi(x)\phi(y)]$, for all $x, y \in L$.
- ϕ is called a ***monomorphism*** if $\mathrm{Ker}\phi = 0$, an ***epimorphism*** if $\mathrm{Im}\phi = L'$, an ***isomorphism*** if it is both mono- and epi-.